

2026



IP Network Basics

B2 - BACnet / Modbus Advanced - IT Basics

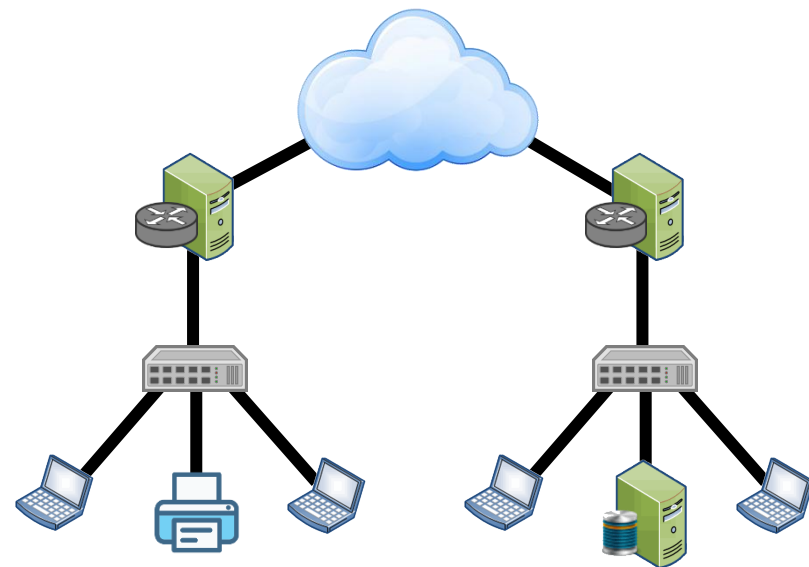
- Network & Components
- Network Services
- IP-Address
- MAC-Address
- Ports
- Network models

- IP stands for Internet Protocol
- The Internet Protocol is a family of protocols (IP, ICMP, IGMP, TCP, UDP)
- The Internet Protocol is currently available in version 4 and version 6
- In this presentation we focus on IPv4 based networks, however, some of the terms for components, services, etc. are being used in a similar way on non-IP networks

Network & Components

What is a network

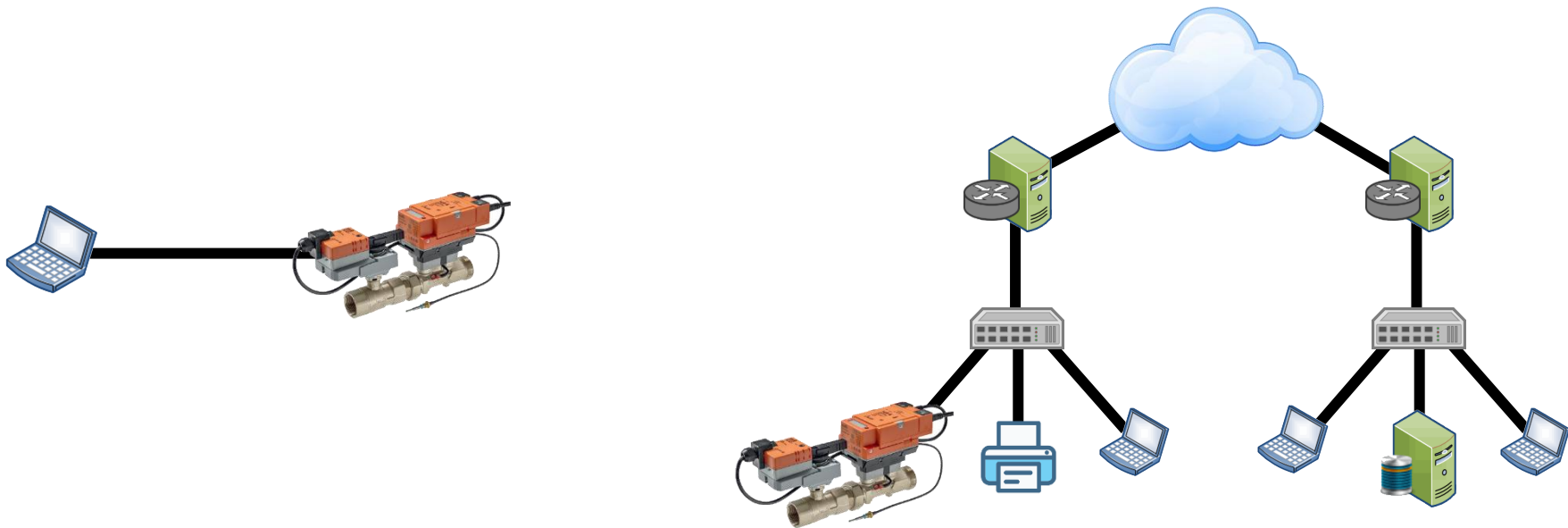
- A network consists of two to several million devices
- Sharing resources (printers, memory, etc.)
- Services (communication, IoT, etc.)
- Internet is the largest existing network and is a network of private and public networks



What is a network

Example EnergyValve

- A network consists of two to several million devices
- Sharing resources (printers, memory, etc.)
- Services (communication, IoT, etc.)
- Internet is the largest existing network and is a network of private and public networks



Client and Server

- All end devices in the network are classified as hosts
- Servers provide services (e.g., web server)
- Client use services (e.g., web browser)



Webserver



Mailserver



Fileserver



Tablet



Computer



Printer

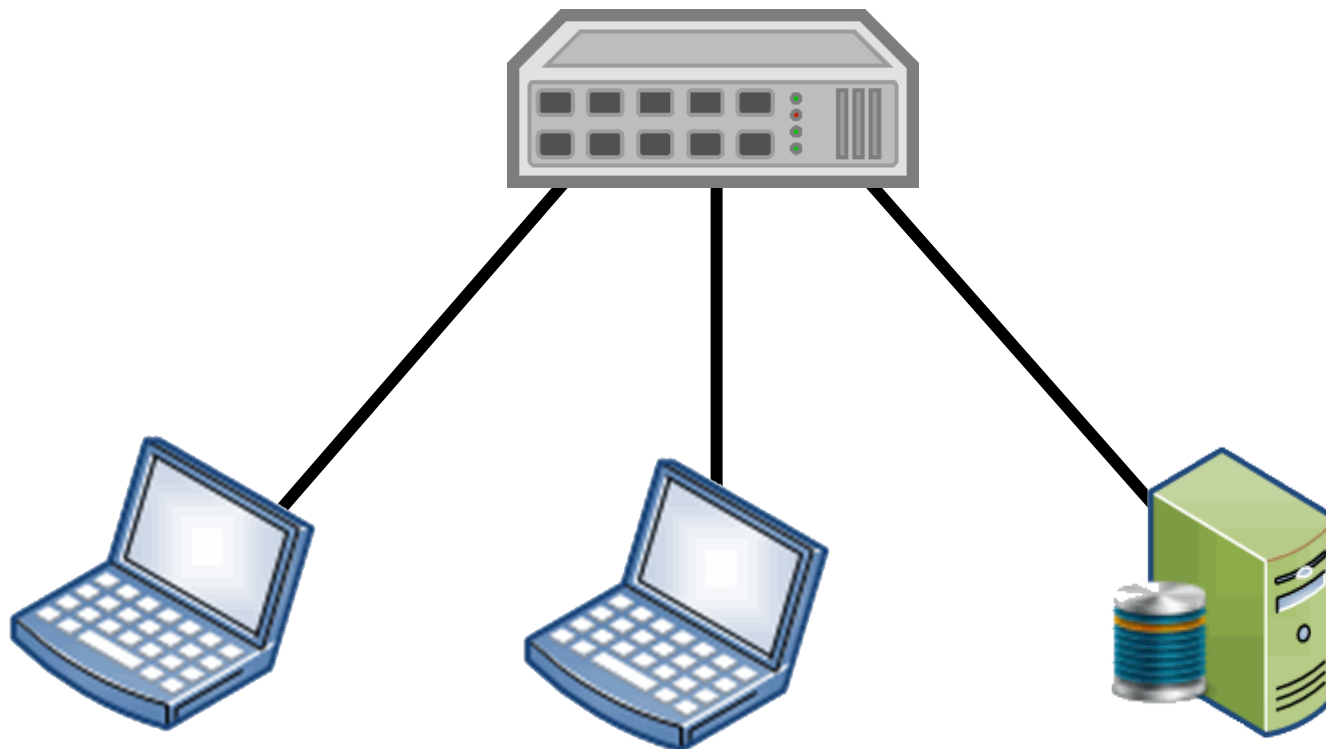


Laptop



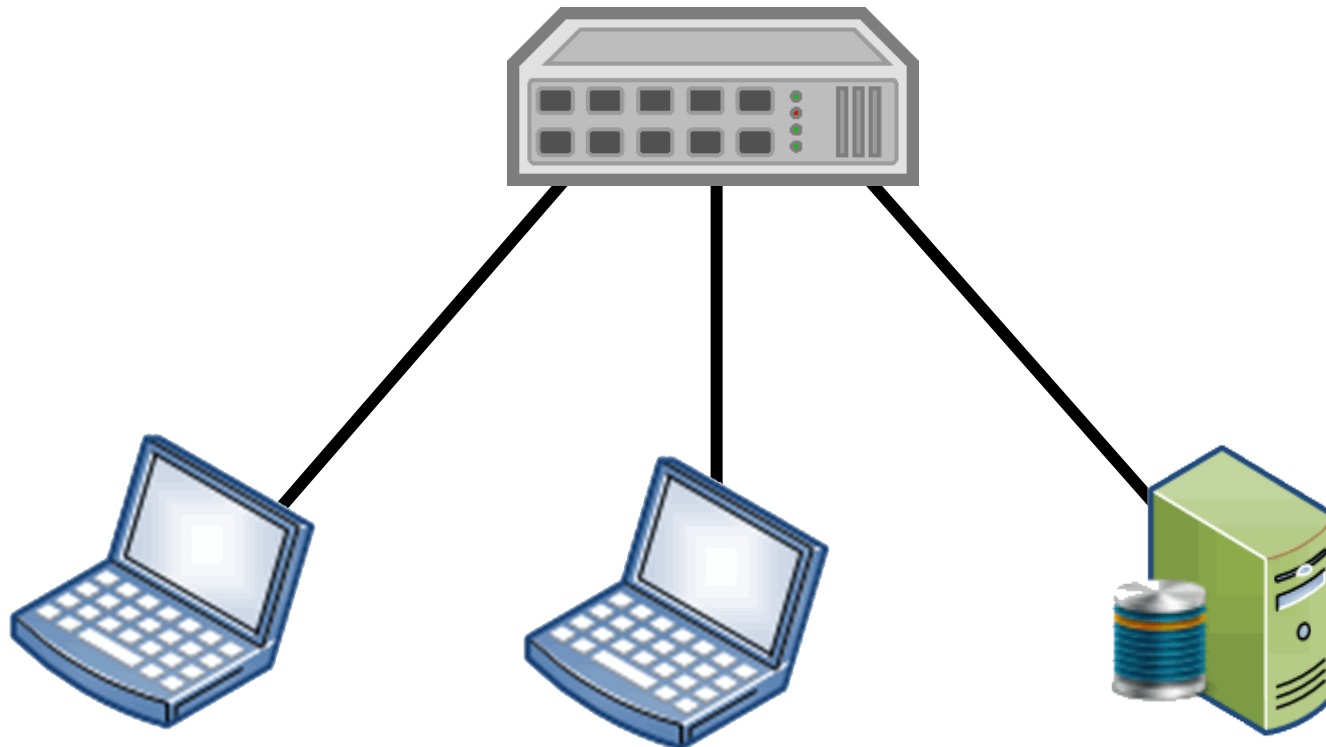
Hub (out of date)

- Connects end devices to each other
- Sends all incoming frames to all other devices



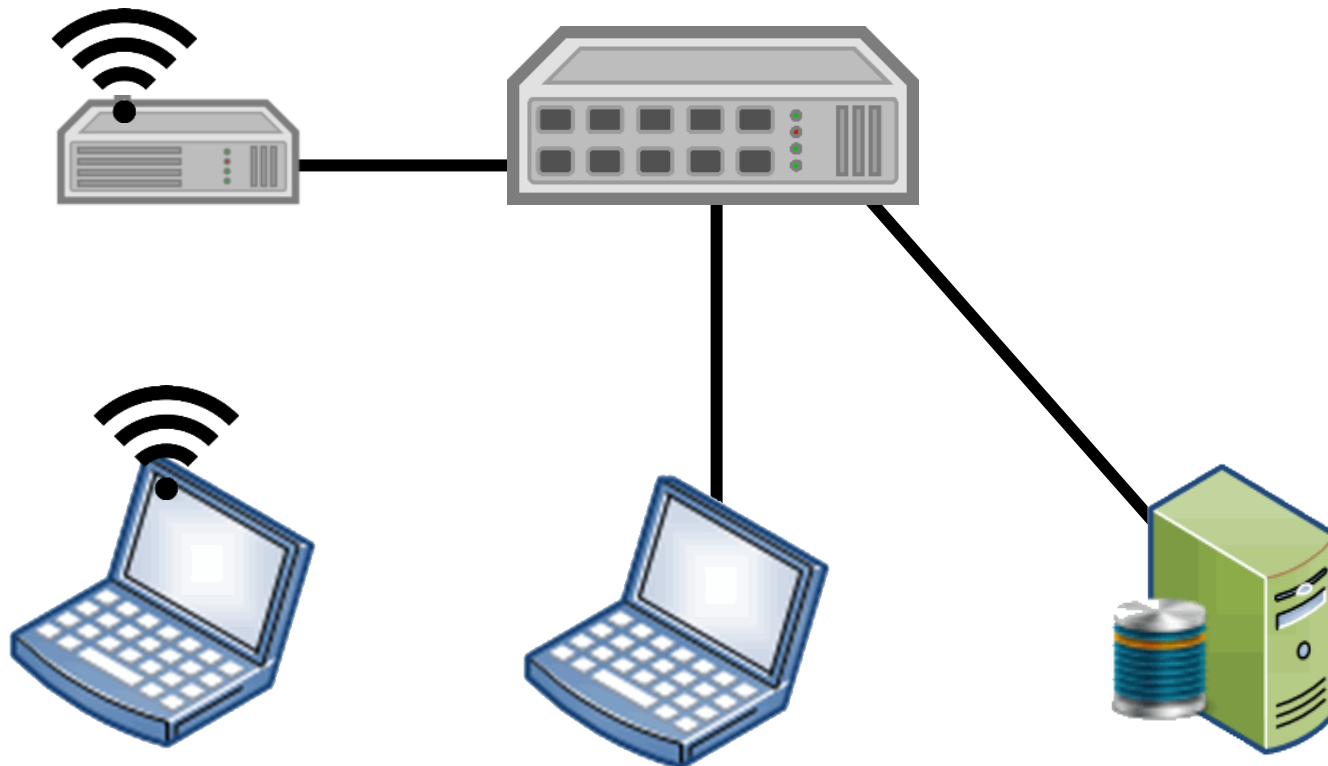
Switch

- Connects end devices to each other
- Forwards incoming frames to the device addressed



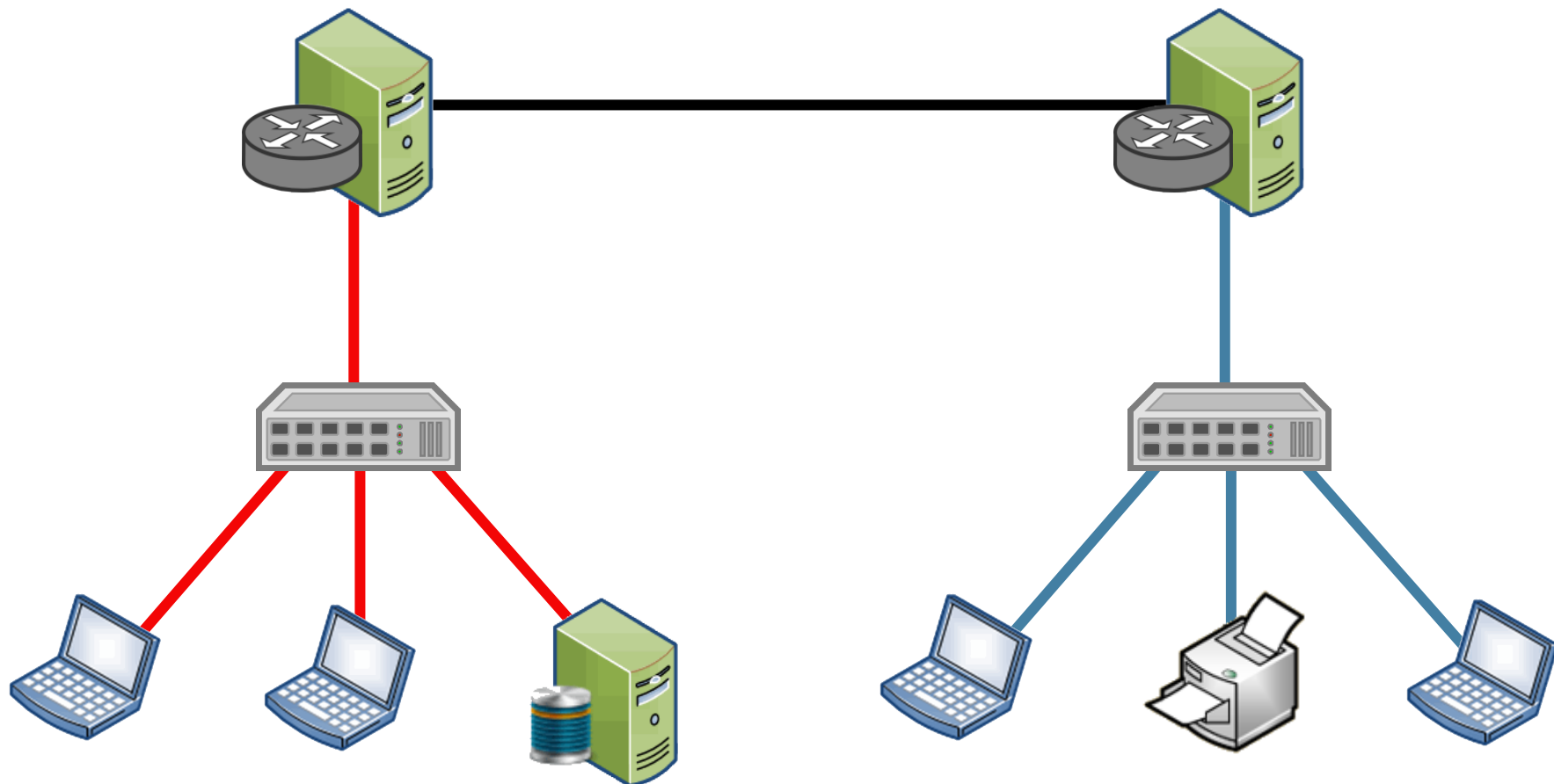
Wi-Fi Access Point

- WLAN



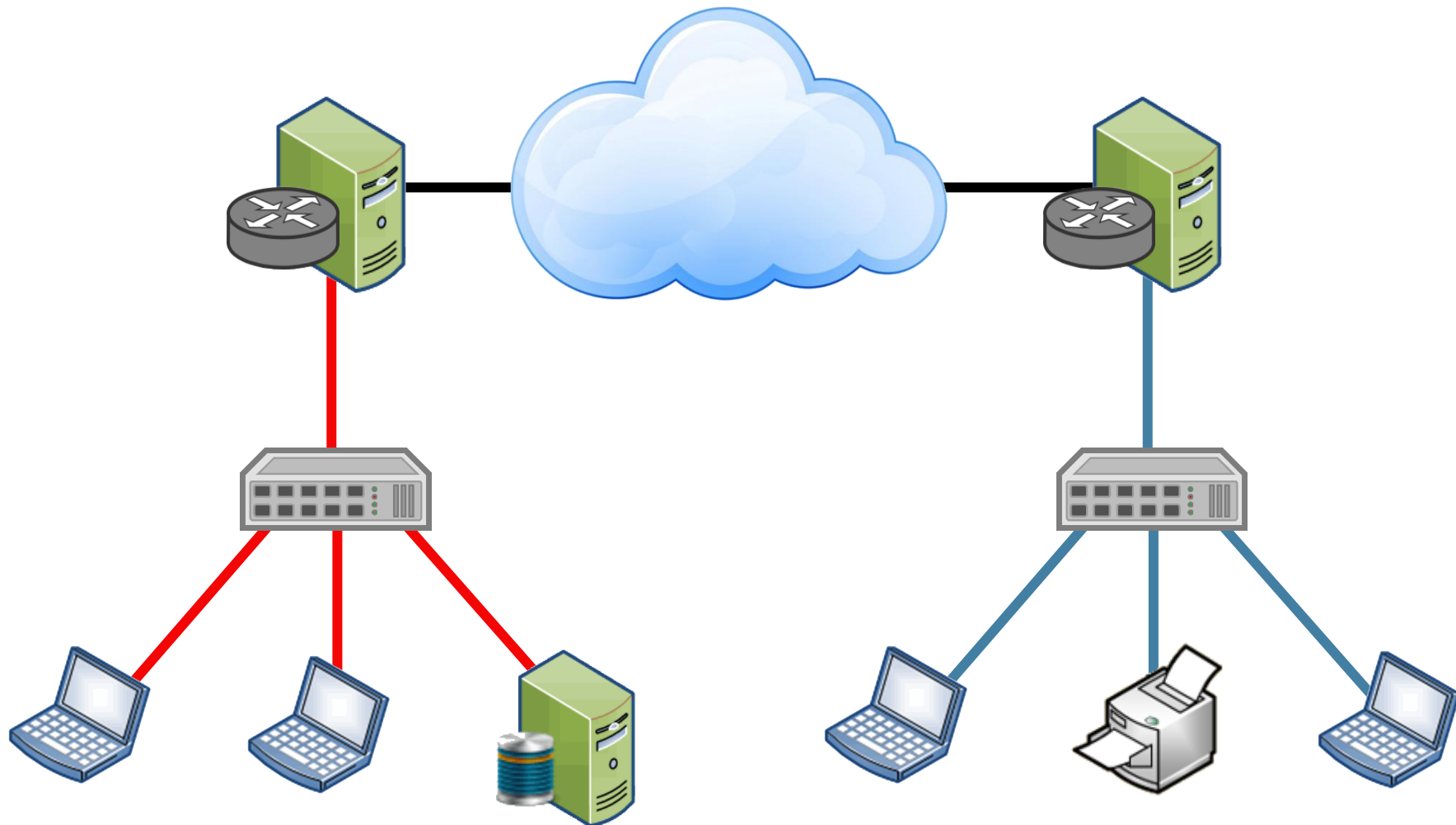
Router

- Connects networks by routing data packets to other networks



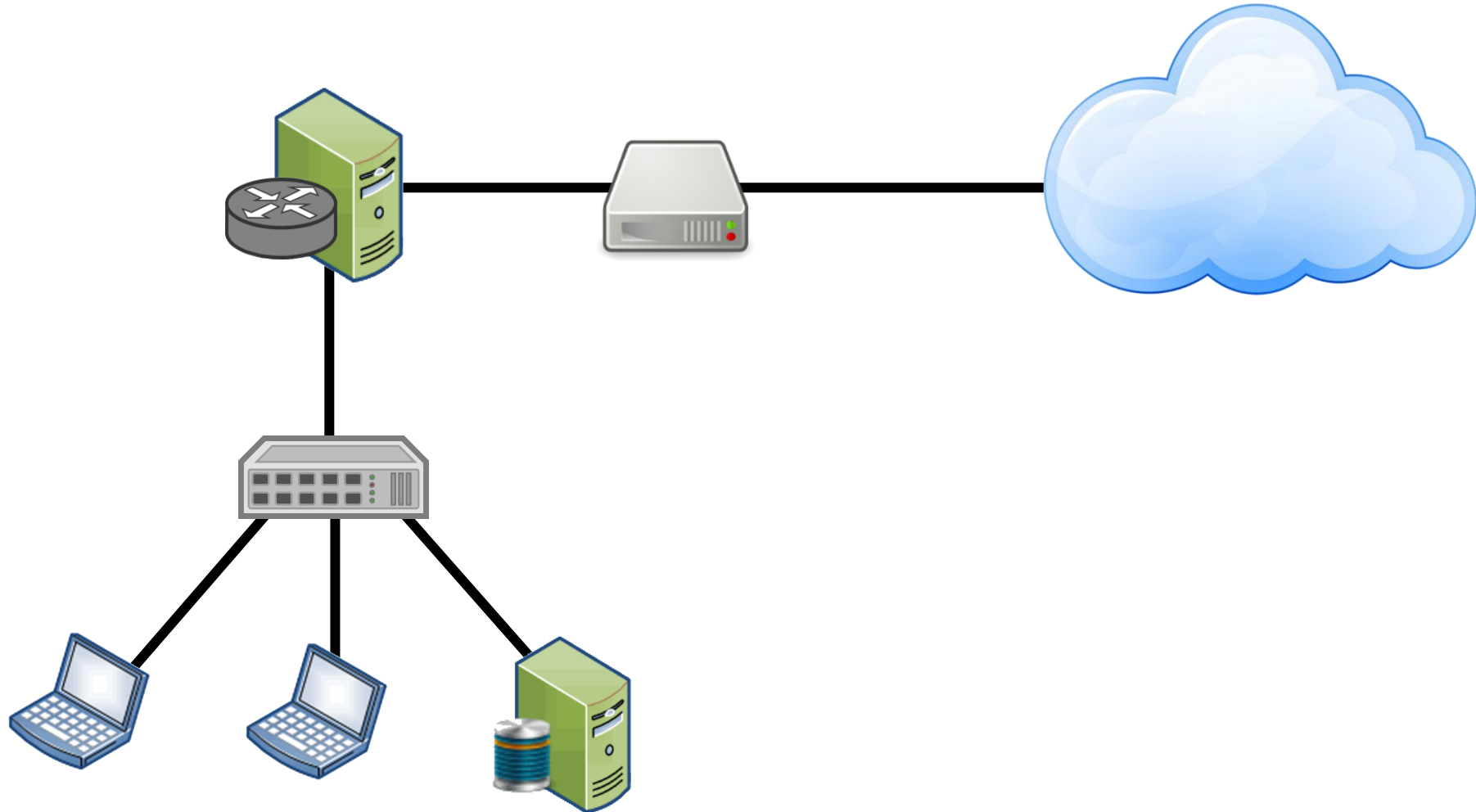
Router

- Connects networks by routing data packets to other networks



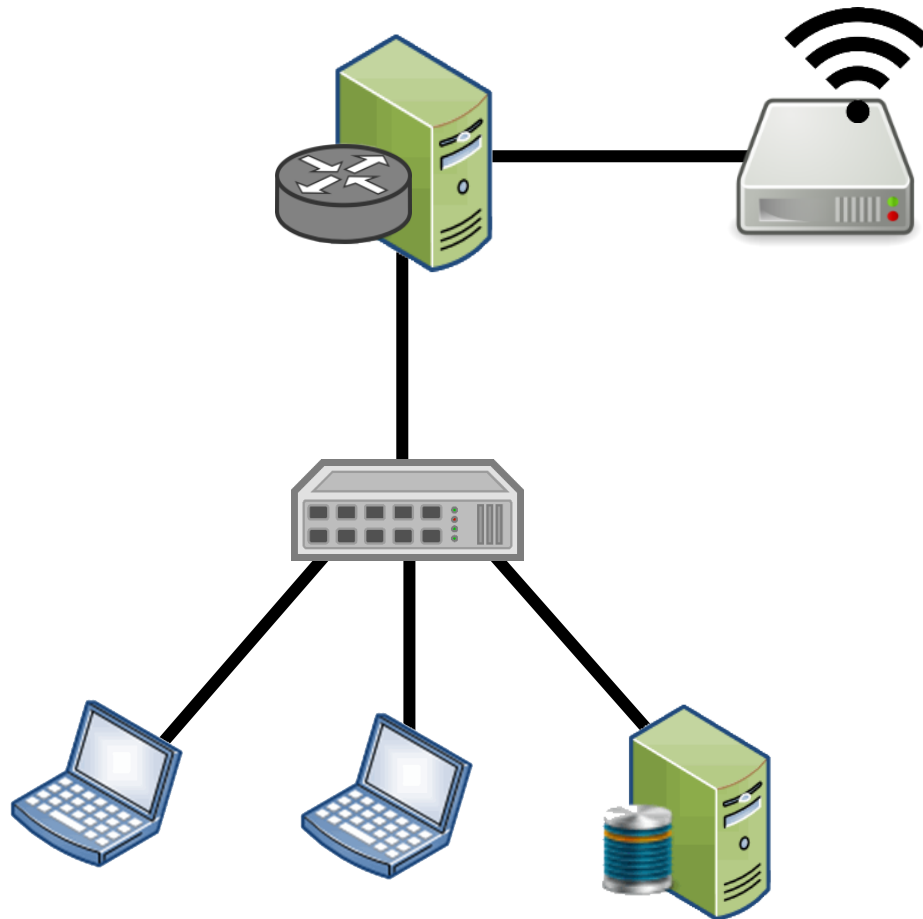
Modem

- Signal conversion
 - Ethernet <-> Fiberglass, Copper Cable (e.g., ISDN, Coax)



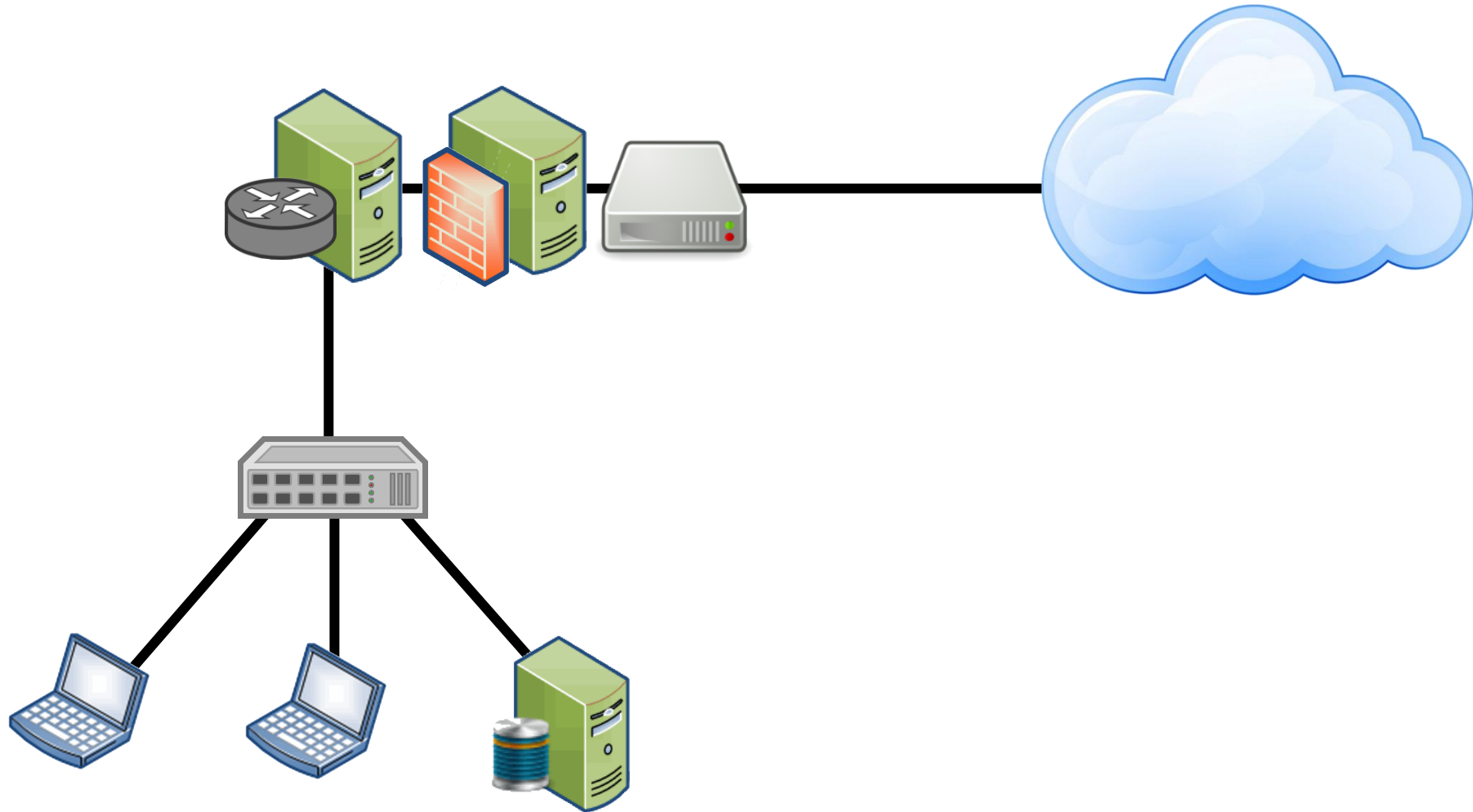
Modem

- Signal conversion
 - Ethernet <-> Mobile network



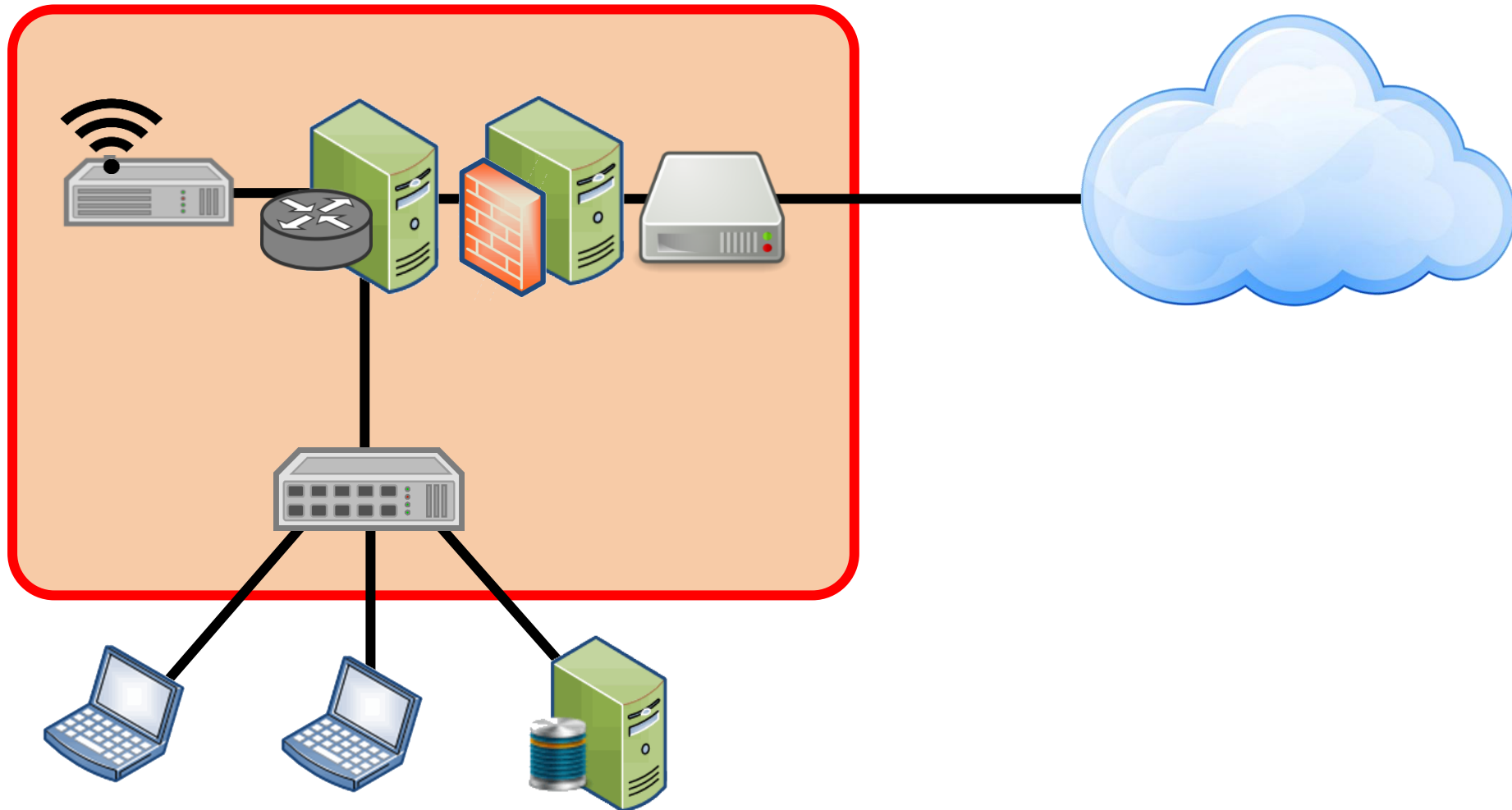
Firewall

- Serves security
- Decides which packets are forwarded based on defined rules



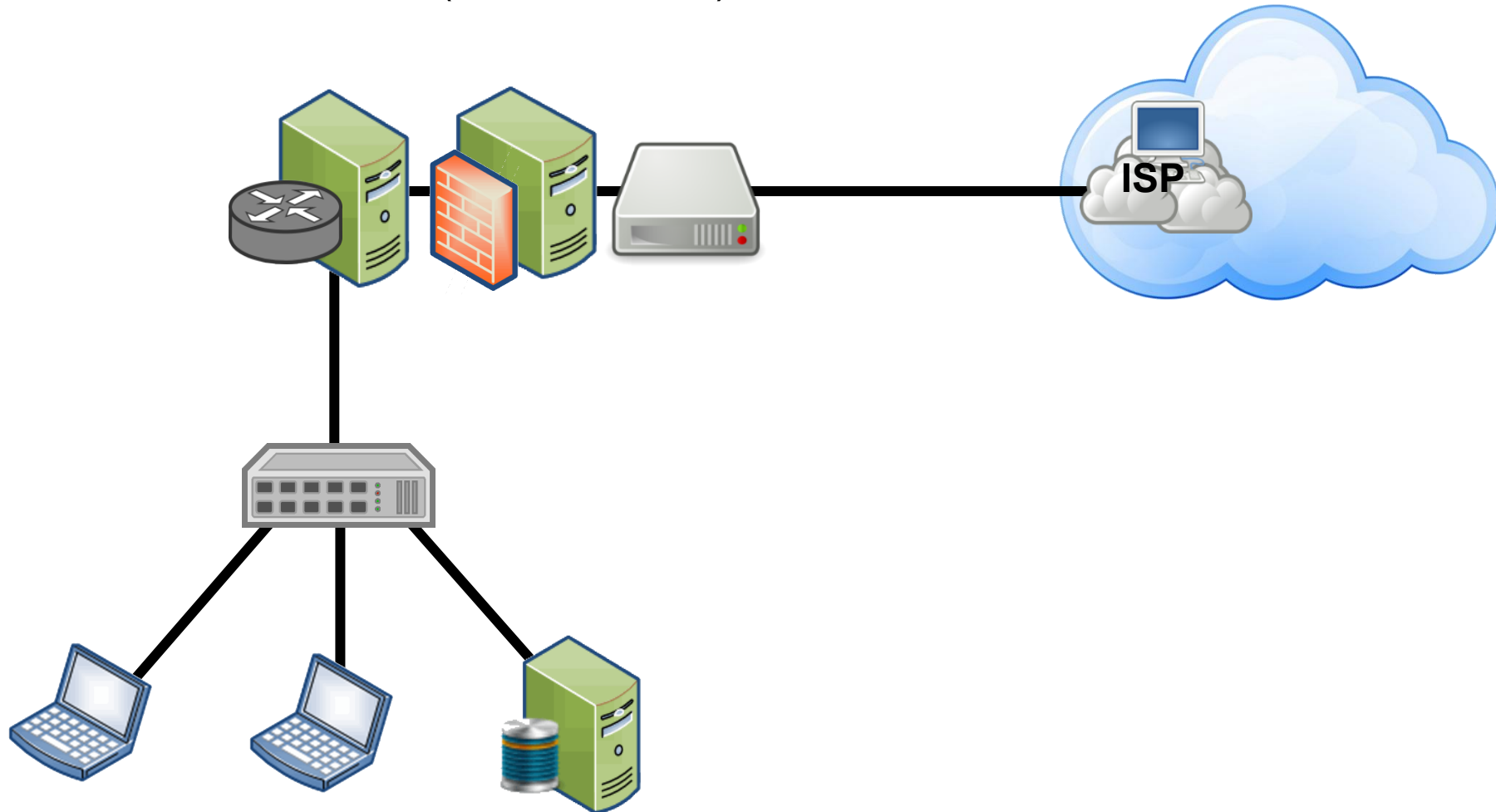
Integrated Access Device (IAD)

- Combination of various network components



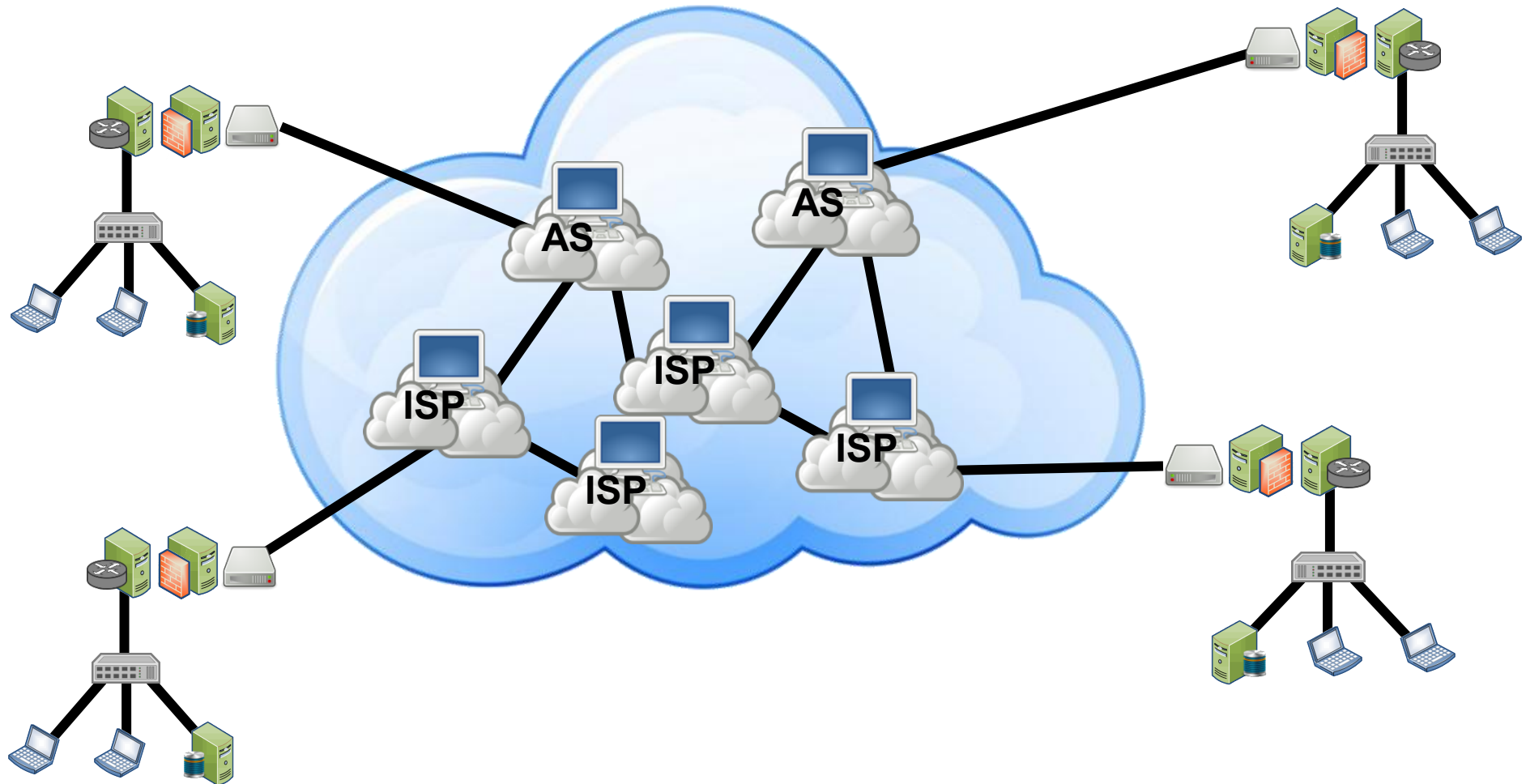
Internet Service Provider (ISP)

- Service provider for connection to a WAN
- Allocates IP addresses (DHCP server)



Internet

- Networking of individual entities ISP and AS (autonomous system)



IP Network Services

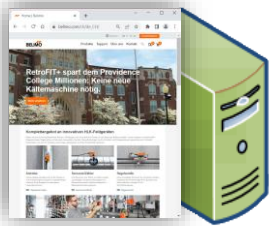
A service is software that provides a function



A server is a network device with services installed. Servers are typically dedicated computers in the network meant for service provisioning. However, you can also install a service on your notebook, which typically is a client, making it a server.



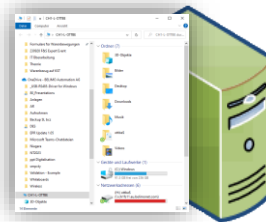
Examples of Network Services



Webserver



Mailserver



Fileserver



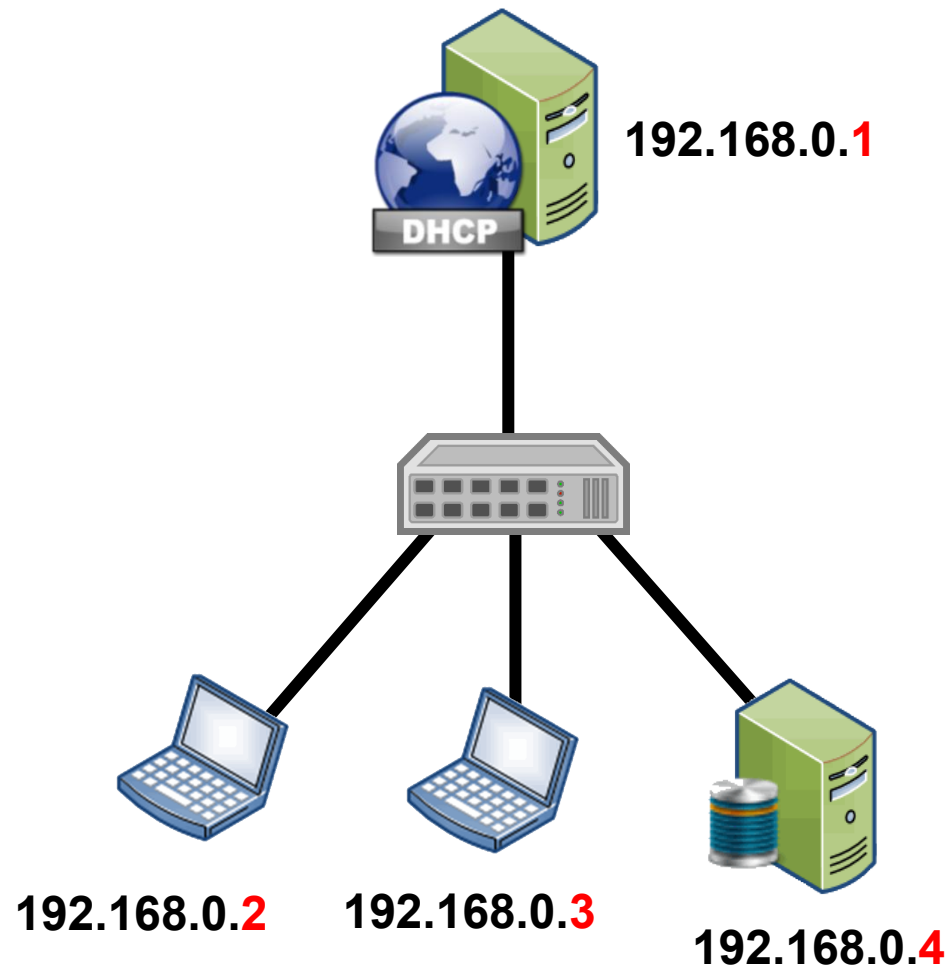
DHCP



DNS

DHCP

- Service to automatically allocate a unique address to network components
- The DHCP service distributes a unique IPv4 address, subnet mask, and default gateway to each client.



Network settings on your laptop

The screenshot shows the 'Allgemein' tab of the 'Eigenschaften von Internetprotokoll, Version 4 (TCP/IPv4)' window. The text at the top states: 'IP-Einstellungen können automatisch zugewiesen werden, wenn das Netzwerk diese Funktion unterstützt. Wenden Sie sich andernfalls an den Netzwerkadministrator, um die geeigneten IP-Einstellungen zu beziehen.' Below this, the radio button 'IP-Adresse automatisch beziehen' is selected. The 'Folgende IP-Adresse verwenden' section is disabled. The 'DNS-Serveradresse automatisch beziehen' radio button is also selected, and the 'Folgende DNS-Serveradressen verwenden' section is disabled. The 'Einstellungen beim Beenden überprüfen' checkbox is unchecked. The 'Erweitert...' button is visible at the bottom right.

Dynamic (DHCP):

The device acts as a DHCP client and requests an IP address and all network parameters from the DHCP server

The screenshot shows the 'Allgemein' tab of the 'Eigenschaften von Internetprotokoll, Version 4 (TCP/IPv4)' window. The text at the top is the same as the DHCP screenshot. In this configuration, the radio button 'Folgende IP-Adresse verwenden' is selected. The 'IP-Adresse' field is set to '192 . 168 . 0 . 11', the 'Subnetzmaske' is '255 . 255 . 255 . 0', and the 'Standardgateway' is '192 . 168 . 0 . 1'. The 'DNS-Serveradresse automatisch beziehen' radio button is unselected, and the 'Folgende DNS-Serveradressen verwenden' section is active, with 'Bevorzugter DNS-Server' set to '10 . 10 . 100 . 92' and 'Alternativer DNS-Server' set to '10 . 10 . 100 . 90'. The 'Einstellungen beim Beenden überprüfen' checkbox is unchecked. The 'Erweitert...' button is visible at the bottom right.

Static:

IP address and network parameters must be set manually. Must match the router settings (IP range, subnet mask, etc.).

Network settings on EnergyValve

Network Configuration

50:2d:f4:0f:df:67 MAC Address

80 HTTP Port

443 HTTPS Port

☒ DHCP/Zeroconf
☐ Static/Zeroconf

192.168.0.10 IP Address

255.255.255.0 Network Mask

192.168.0.1 Gateway

8.8.8.8 DNS Nameserver 1

8.8.4.4 DNS Nameserver 2

10.20.255.255 Broadcast Address

169.254.132.178 ZeroConf Address

Change IP Configuration

Network Configuration

50:2d:f4:0f:df:67 MAC Address

80 HTTP Port

443 HTTPS Port

☐ DHCP/Zeroconf
☒ Static/Zeroconf

192.168.0.10 IP Address

255.255.255.0 Network Mask

192.168.0.1 Gateway

8.8.8.8 DNS Nameserver 1

8.8.4.4 DNS Nameserver 2

10.20.255.255 Broadcast Address

169.254.132.178 ZeroConf Address

Change IP Configuration

Dynamic (DHCP):

The device acts as a DHCP client and requests an IP address and all network parameters from the DHCP server

Static:

IP address and network parameters must be set manually. Must match the router settings (IP range, subnet mask, etc.).

Peer-to-Peer with no DHCP server Static



Internetprotokoll, Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 0 . X

Subnet mask: 255 . 255 . 255 . 0

Default gateway: . . .

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

Network Configuration

50:2d:f4:0f:df:67 MAC Address

80 HTTP Port

443 HTTPS Port

☐ DHCP/Zeroconf

☒ Static/Zeroconf

192.168.0.10 IP Address

255.255.255.0 Network Mask

192.168.0.1 Gateway

8.8.8.8 DNS Nameserver 1

8.8.4.4 DNS Nameserver 2

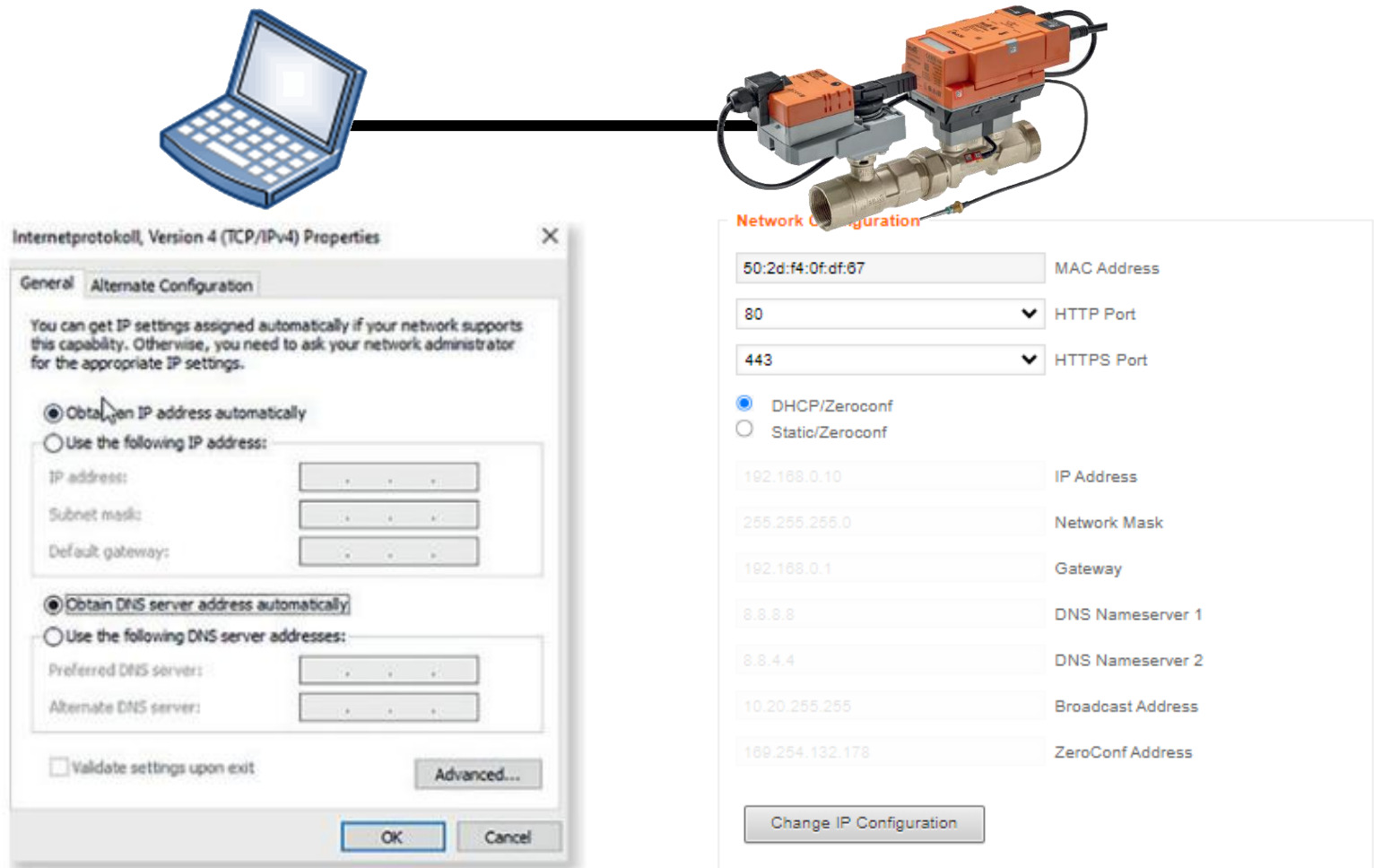
10.20.255.255 Broadcast Address

169.254.132.178 ZeroConf Address

Change IP Configuration

For more info see [Web Server Manual](#) EnergyValve

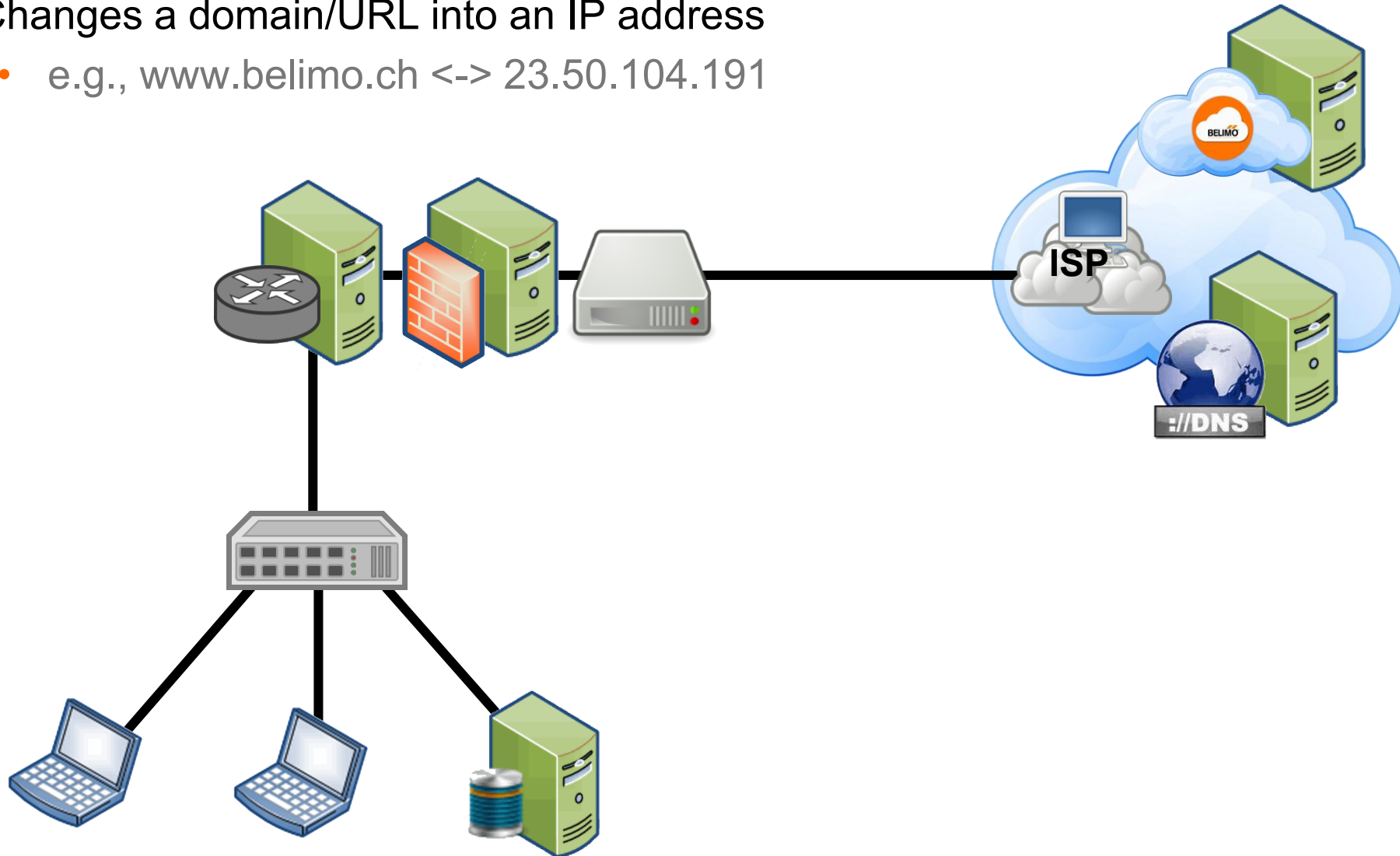
Peer-to-Peer with no DHCP server Dynamic (Zeroconfig)



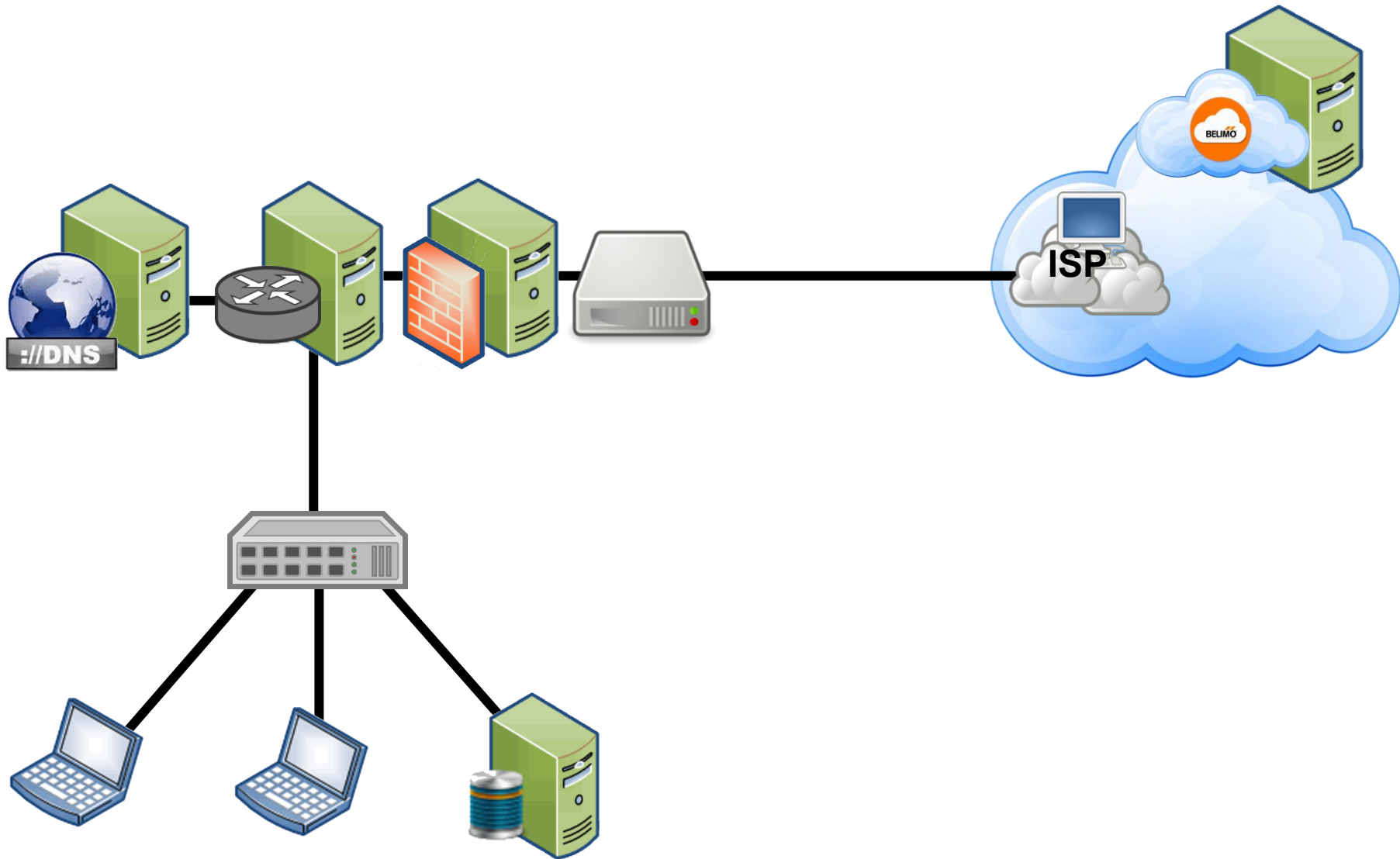
For more info see [Web Server Manual](#) EnergyValve

DNS

- Changes a domain/URL into an IP address
 - e.g., www.belimo.ch <-> 23.50.104.191



DNS - Local



IP Addresses

IP address structure (IPv4)

- Dotted Decimal Notation x.x.x.x
 - Each octet (x) has 8 bits -> range 0-255
- In theory there are $2^{32} = 4'294'967'296$ IP-Addresses
- IP address is divided into a net ID and a host ID using the subnet mask
- Example:

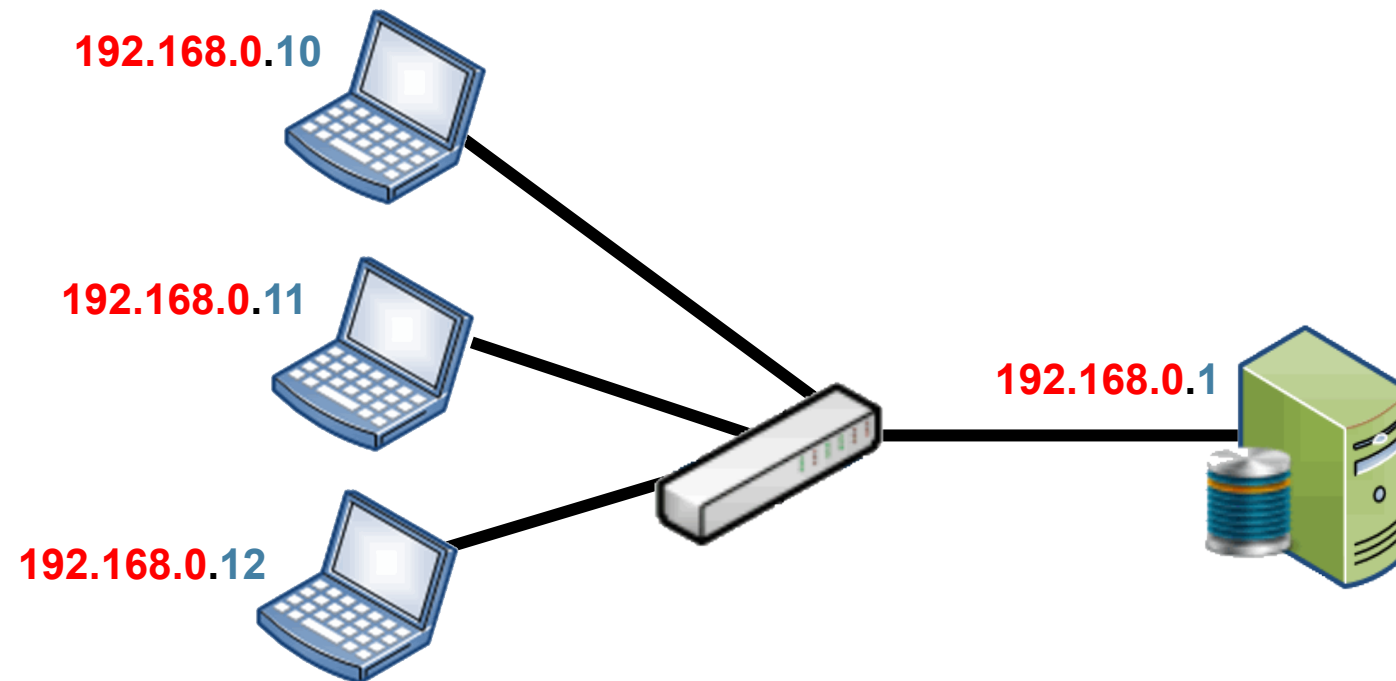
IP-Address:	192.168. 0 .10	1100'0000.1010'1000.1011'0010.0111'1101
Subnet mask:	255.255.255. 0	1111'1111. 1111'1111 . 1111'1111.0000'0000
IP-Address:	192.168. 0 .10	1100'0000.1010'1000.1011'0010.0111'1101
	Network part Host part	

Network ID: 192.168.0

Host ID: 10

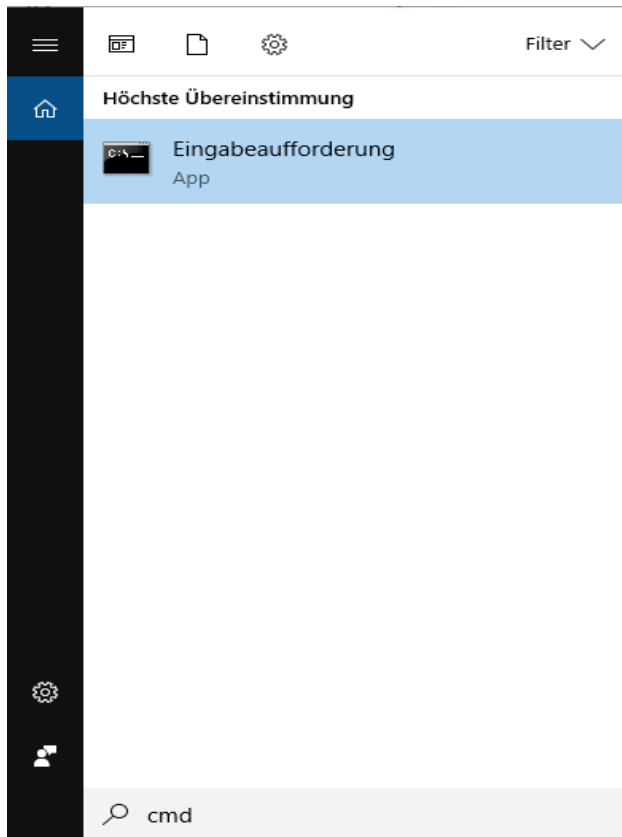
The network includes IP-Addresses: 192.168.0.1 – 192.168.0.254

Net ID and Host ID



IP-Address & Subnet mask

Check your IP Settings
-> *ipconfig*



```
Eingabeaufforderung
Windows-IP-Konfiguration

Ethernet-Adapter Ethernet:

    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix: eu.belimonet.com

Drahtlos-LAN-Adapter LAN-Verbindung* 1:

    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix:

Drahtlos-LAN-Adapter LAN-Verbindung* 2:

    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix:

Ethernet-Adapter Ethernet 2:

    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix:

Drahtlos-LAN-Adapter WLAN:

    Verbindungsspezifisches DNS-Suffix: home
    IPv6-Adresse. . . . . : 2a02:21b4:9873:4300:b9f5:f8a5:1fa0:938a
    Temporäre IPv6-Adresse. . . . . : 2a02:21b4:9873:4300:a95b:503c:758:cd17
    Verbindungslokale IPv6-Adresse . . : fe80::f3ac:8b49:131e:b417%13
    IPv4-Adresse . . . . . : 192.168.0.18
    Subnetzmaske . . . . . : 255.255.255.0
    Standardgateway . . . . . : fe80::7a29:edff:fe3d:b8c0%13
                                192.168.0.254

Ethernet-Adapter Bluetooth-Netzwerkverbindung:

    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix:

C:\Users\ottbe>
```

IP-Address (IPv4)

- Public (global) & private IP addresses
 - Public IP addresses
 - are administered/allocated by IANA (Internet Assigned Numbers Authority).
 - are unique worldwide
 - are directly accessible on the Internet
 - Private IP addresses
 - are mostly managed/assigned locally
-
- Loopback
 - 127.0.0.0/8
 - RFC1918 - Private Address Space
 - 10.0.0.0/8, 172.16.0.0/12 und 192.168.0.0/16
 - RFC3927 - APIPA / Link-Local
 - 169.254.0.0/16
 - RFC6598 - Shared Address Space (CGN)
 - 100.64.0.0/12
 - RFC5737 - Documentation prefixes
 - 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24
 - RFC8190 - Special-Purpose IP Address Registries
 - Complete list of special prefixes



Private IP-Address

```
Eingabeaufforderung
Windows-IP-Konfiguration

Ethernet-Adapter Ethernet:

    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix: eu.belimonet.com

Drahtlos-LAN-Adapter LAN-Verbindung* 1:

    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix:

Drahtlos-LAN-Adapter LAN-Verbindung* 2:

    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix:

Ethernet-Adapter Ethernet 2:

    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix:

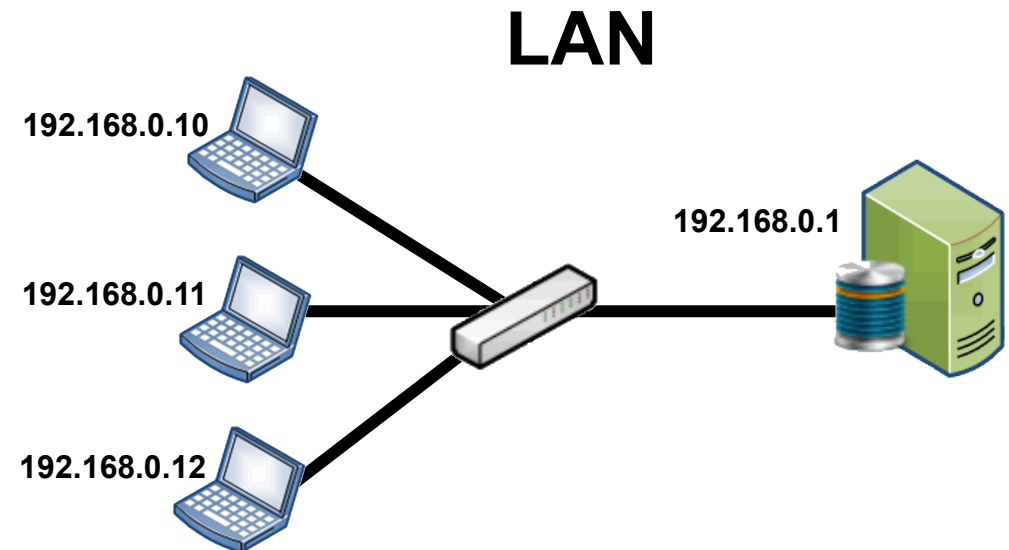
Drahtlos-LAN-Adapter WLAN:

    Verbindungsspezifisches DNS-Suffix: home
    IPv6-Adresse. . . . . : 2a02:21b4:9873:4300:b9f5:f8a5:1fa0:938a
    Temporäre IPv6-Adresse. . . . . : 2a02:21b4:9873:4300:a95b:503c:758:cd17
    Verbindungslokale IPv6-Adresse. . . . . : fe80::198c:8843:131e:7%13
    IPv4-Adresse. . . . . : 192.168.0.18
    Subnetzmaske. . . . . : 255.255.255.0
    Standardgateway. . . . . : 192.168.0.254

Ethernet-Adapter Bluetooth-Netzwerkverbindung:

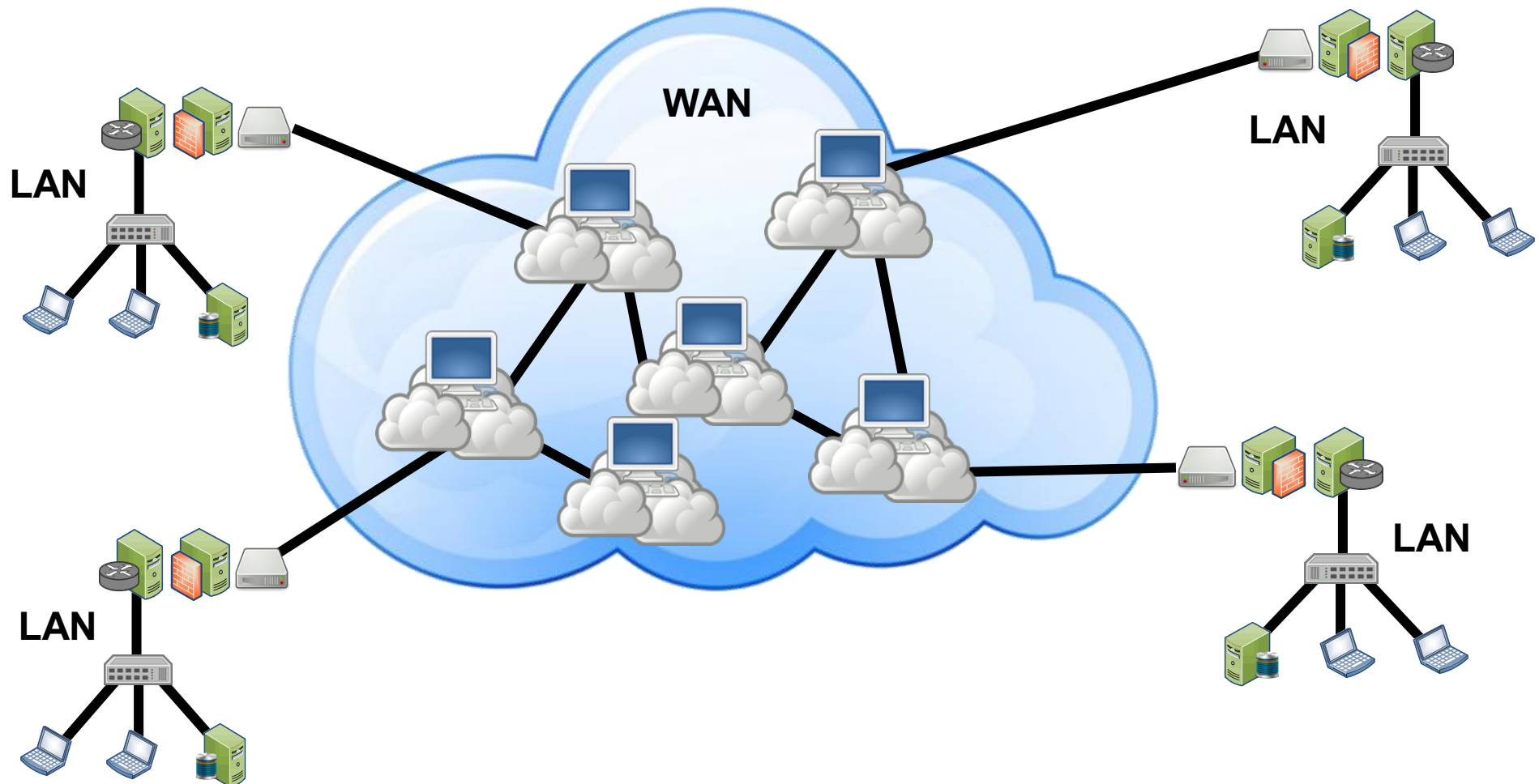
    Medienstatus. . . . . : Medium getrennt
    Verbindungsspezifisches DNS-Suffix:

C:\Users\ottbe>
```



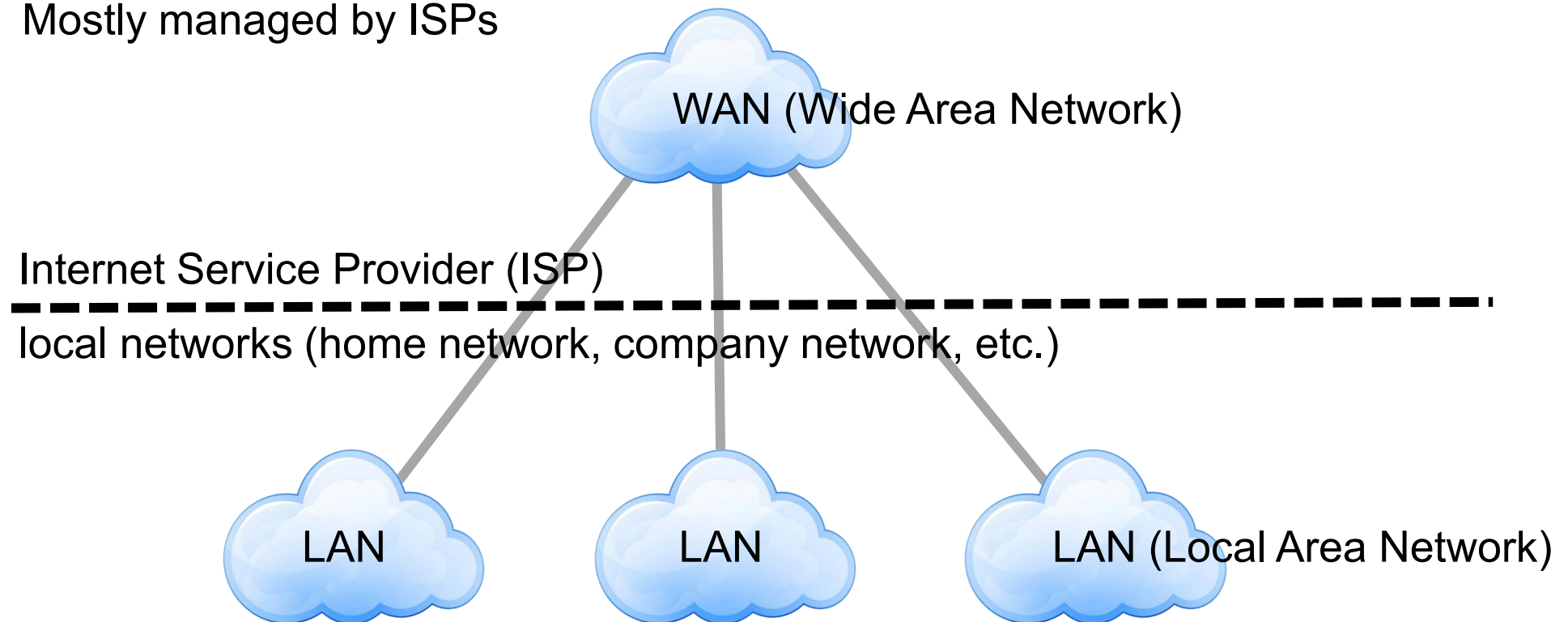
Limited number of IP-Addresses

4'294'967'296 IP-Addresses are not enough! How do we solve it?



Network infrastructures

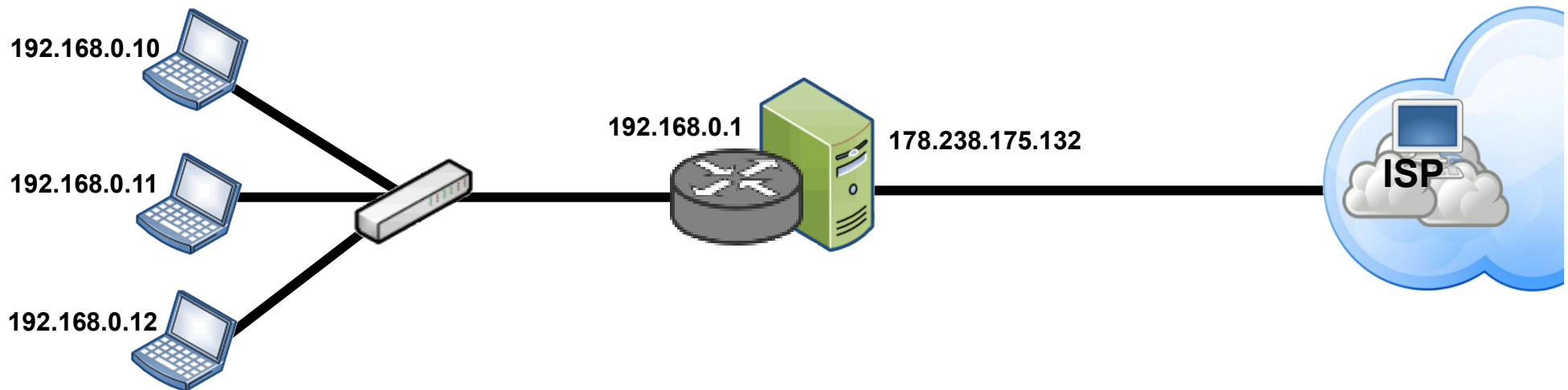
- Long distances
- LAN spanning
- Mostly managed by ISPs



- short distances
- e.g., home network, company network; often managed locally

Network Address Translation NAT

- The computers in a network share a public IP address
- The router translates each request to the Internet and changes the IP address to the outside
- An internal table in the router allows the translation back to the original address when the response arrives.



Public IP-Address

My IP Address is:

IPv6: ? [2a02:21b4:9873:4300:a95b:503c:758:cd17](#)

IPv4: ? [178.238.175.132](#)

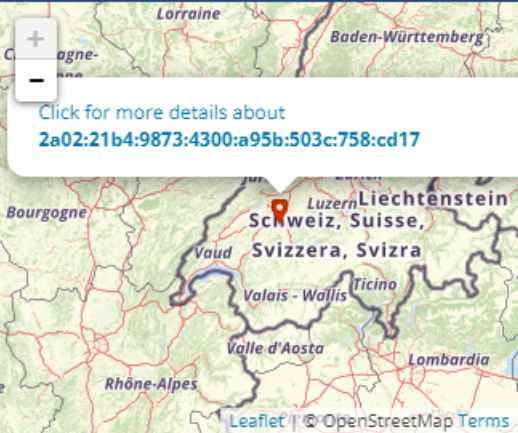
My IP Information:

ISP:	Swisscom AG
City:	Bern
Region:	Bern
Country:	Switzerland

Your location may be exposed!

HIDE MY IP ADDRESS NOW

[Show Complete IP Details](#)

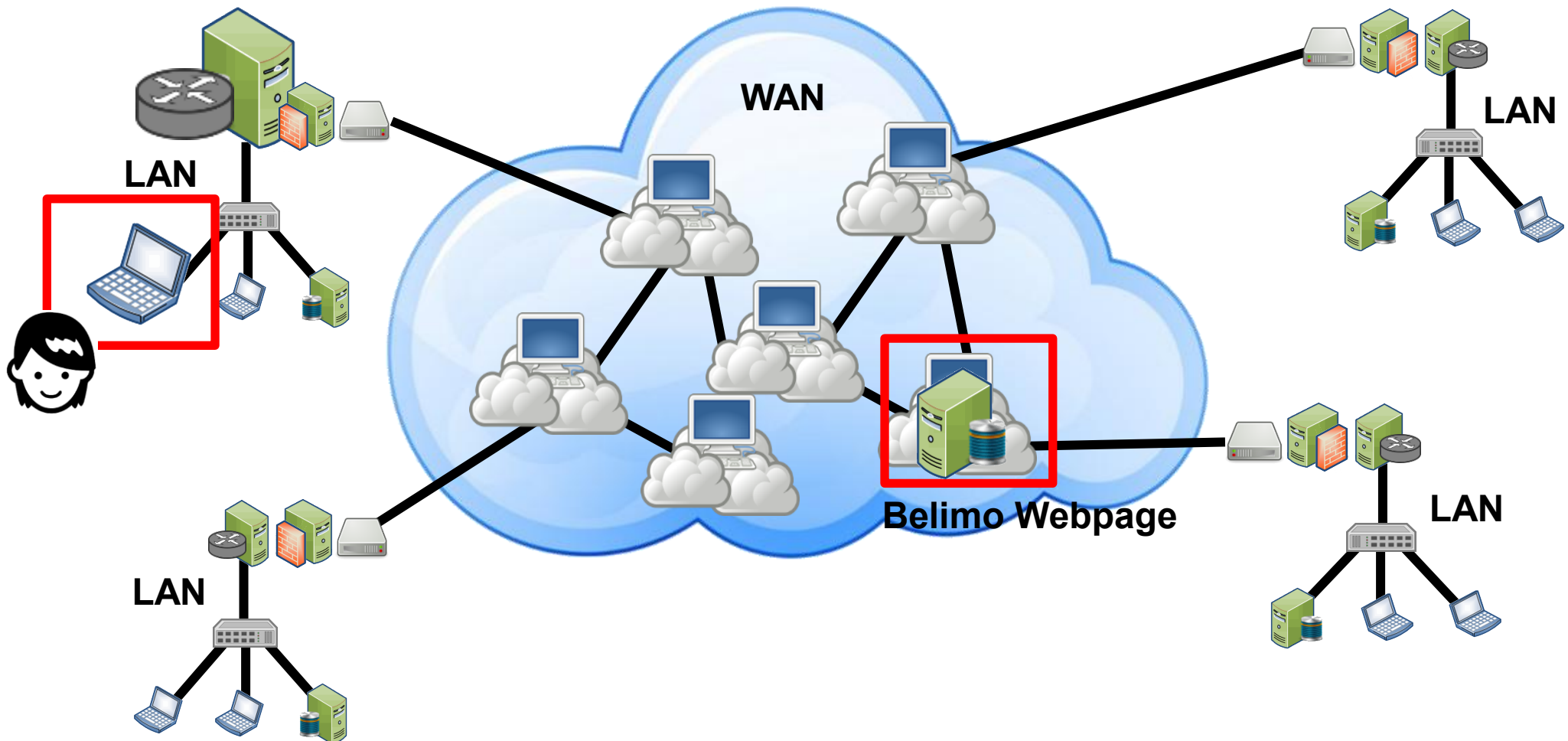


Location not accurate?
[Update My IP Location](#)

Check:

<https://whatismyipaddress.com>

Example Belimo Webpage



Example Belimo Webpage



Check route to www.belimo.ch
-> *tracert* www.belimo.ch

The screenshot shows a Windows interface. On the left is a sidebar with a search bar and icons for 'Höchste Übereinstimmung' and 'Eingabeaufforderung App'. The main area displays a terminal window with the following output:

```
C:\Users\ottbe>tracert www.belimo.ch

Routenverfolgung zu e13686.b.akamaiedge.net [23.12.134.12]
über maximal 30 Hops:

 1    3 ms    3 ms    4 ms  192.168.0.1
 2   16 ms   15 ms   11 ms  46-127-132-1.dynamic.hispeed.ch [46.127.132.1]
 3   11 ms   14 ms   14 ms  217-168-53-29.static.cablecom.ch [217.168.53.29]
 4   27 ms   31 ms   16 ms  ch-zrh03a-rc1-ae50-0.aorta.net [84.116.200.241]
 5   21 ms   51 ms   13 ms  ch-zrh01b-ra1-ae1-0.aorta.net [84.116.134.142]
 6   47 ms   15 ms   14 ms  213.46.171.222
 7   28 ms   23 ms   22 ms  mno-b2-link.telia.net [62.115.116.126]
 8    *      *      *      Zeitüberschreitung der Anforderung.
 9   18 ms   21 ms   19 ms  a23-12-134-12.deploy.static.akamaitechnologies.com [23.12.134.12]
```

MAC Address

Network Addresses

- Network addresses serve as unique identifiers in the network
- Each device in the network has two different addresses
 - Physical address -> MAC address
 - Logical address -> IP address
- Link between IP and MAC address (ARP)

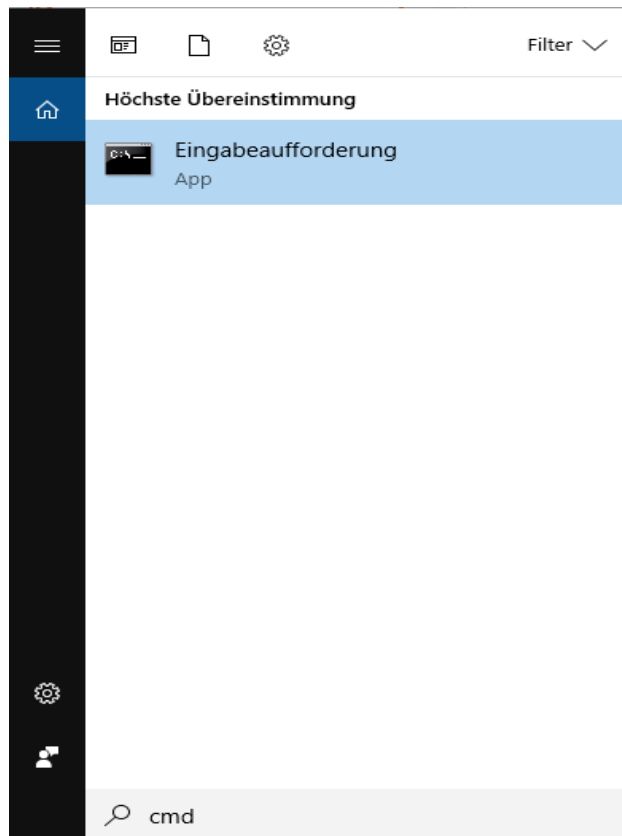
MAC-Address

- **Unique identification**
- **Is allocated by Manufacture of Internet-enabled devices**
- **The MAC address can be used to identify some of the manufacturers.**
- **Address is 48 bit long**
 - Example: 00-16-21-25-AF-D7

ARP list

Check you ARP list

-> arp -a



```
C:\Users\ottbe>arp -a

Schnittstelle: 192.168.0.115 --- 0xd
  Internetadresse   Physische Adresse   Typ
  192.168.0.1       34-2c-c4-68-7e-ec   dynamisch
  192.168.0.255     ff-ff-ff-ff-ff-ff   statisch
  224.0.0.2         01-00-5e-00-00-02   statisch
  224.0.0.22        01-00-5e-00-00-16   statisch
  224.0.0.251       01-00-5e-00-00-fb   statisch
  224.0.0.252       01-00-5e-00-00-fc   statisch
  239.255.255.250   01-00-5e-7f-ff-fa   statisch
  255.255.255.255   ff-ff-ff-ff-ff-ff   statisch

Schnittstelle: 10.9.1.11 --- 0x12
  Internetadresse   Physische Adresse   Typ
  10.9.1.12         00-09-0f-aa-00-02   dynamisch
  224.0.0.2         01-00-5e-00-00-02   statisch
  224.0.0.22        01-00-5e-00-00-16   statisch
  224.0.0.251       01-00-5e-00-00-fb   statisch
  224.0.0.252       01-00-5e-00-00-fc   statisch
  239.255.255.250   01-00-5e-7f-ff-fa   statisch
  255.255.255.255   ff-ff-ff-ff-ff-ff   statisch
```

Port

Port

- Ports are part of every IP communication
- Valid port numbers are 0...65'535
- System Ports
- User Ports
- Dynamic Ports

Port	TCP	UDP	Beschreibung	Status
0		UDP	reserviert	offiziell
1	TCP	UDP	TCP Port Service Multiplexer (TCPMUX)	offiziell
2	TCP	UDP	CompressNET ^[1] Management Utility ^[2]	offiziell
3	TCP	UDP	CompressNET ^[1] Compression Process ^[3]	offiziell
4	TCP	UDP	Mesh Connection Protocol	inoffiziell
5	TCP	UDP	Remote Job Entry	offiziell
6	UDP	UDP	nicht zugewiesen	offiziell
7	TCP	UDP	Echo	offiziell
8	TCP	UDP	nicht zugewiesen	offiziell
9	TCP	UDP	Discard	offiziell
10	TCP	UDP	nicht zugewiesen	offiziell
11	TCP	UDP	Systeminformationen über den (systat-Service ^[4])	offiziell
12	TCP	UDP	LBBP	inoffiziell
13	TCP	UDP	Daytime (RFC 867 ^[5])	offiziell
14	TCP	UDP	nicht zugewiesen	offiziell
15	TCP	UDP	netstat Service ^[4]	inoffiziell
16	TCP	UDP	nicht zugewiesen	offiziell
17	TCP	UDP	Quote of the Day (QOTD) – Protokoll, um eine kurze „Nachricht des Tages“ zu übermitteln	offiziell
18	TCP	UDP	Message Send Protocol – Protokoll, um Textnachrichten zwischen Computern zu übermitteln	offiziell
19	TCP	UDP	Character Generator Protocol (CHARGEN)	offiziell

44818	TCP	UDP	EtherNet/IP (Industrial Ethernet)	offiziell
47808	TCP	UDP	BACnet	offiziell
49000	TCP	UDP	Server für Statusinformationen, SOAP (z. B. bei Fritz!Box)	inoffiziell
49151	TCP	UDP	reserviert	offiziell

Network Models

ISO OSI Model



Application / Presentation / Session layer – Data

z.B. HTTP, FTP, SMTP, POP3, DNS, SSH, Telnet



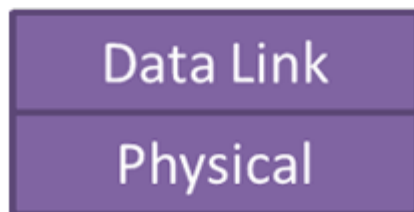
Transport layer – Segment

z.B. TCP, UDP



Network layer – Packet

z.B. IP (IPv4, IPv6), ICMP, IPX



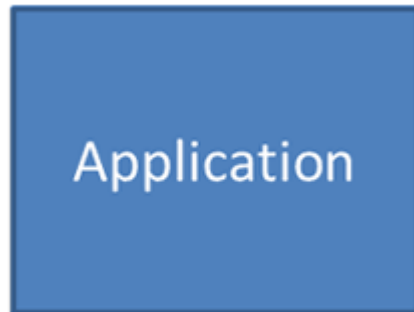
Data Link layer –Frame

z.B. Ethernet, WLAN, ATM, FDDI, PPP, Token Ring

Physical layer – Bit

z.B. 1000BASE-T, Token Ring, ARCNET

TCP/IP Model



Application layer – Application-specific

e.g. HTTP, FTP, SMTP, POP3, DNS, SSH, Telnet



Transport layer – Reliable transmission

e.g. TCP, UDP



Internet layer – Routing

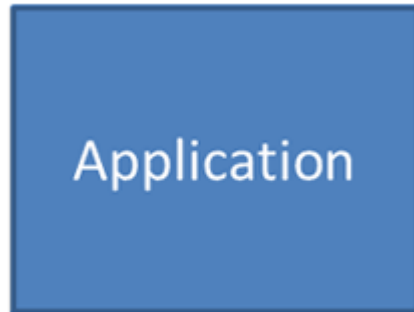
e.g. IP (IPv4, IPv6), ICMP, IPX



Link layer – Data transmission technologies

e.g. Ethernet, WLAN, Coax, Fibre, ATM, FDDI, PPP, Token Ring

Components in ISO OSI Model



Application / Presentation / Session / Transport layer –
Gateway, Content-Switch, Proxy, Layer 4-7-Switch



Network layer –
Router, Layer-3-Switch

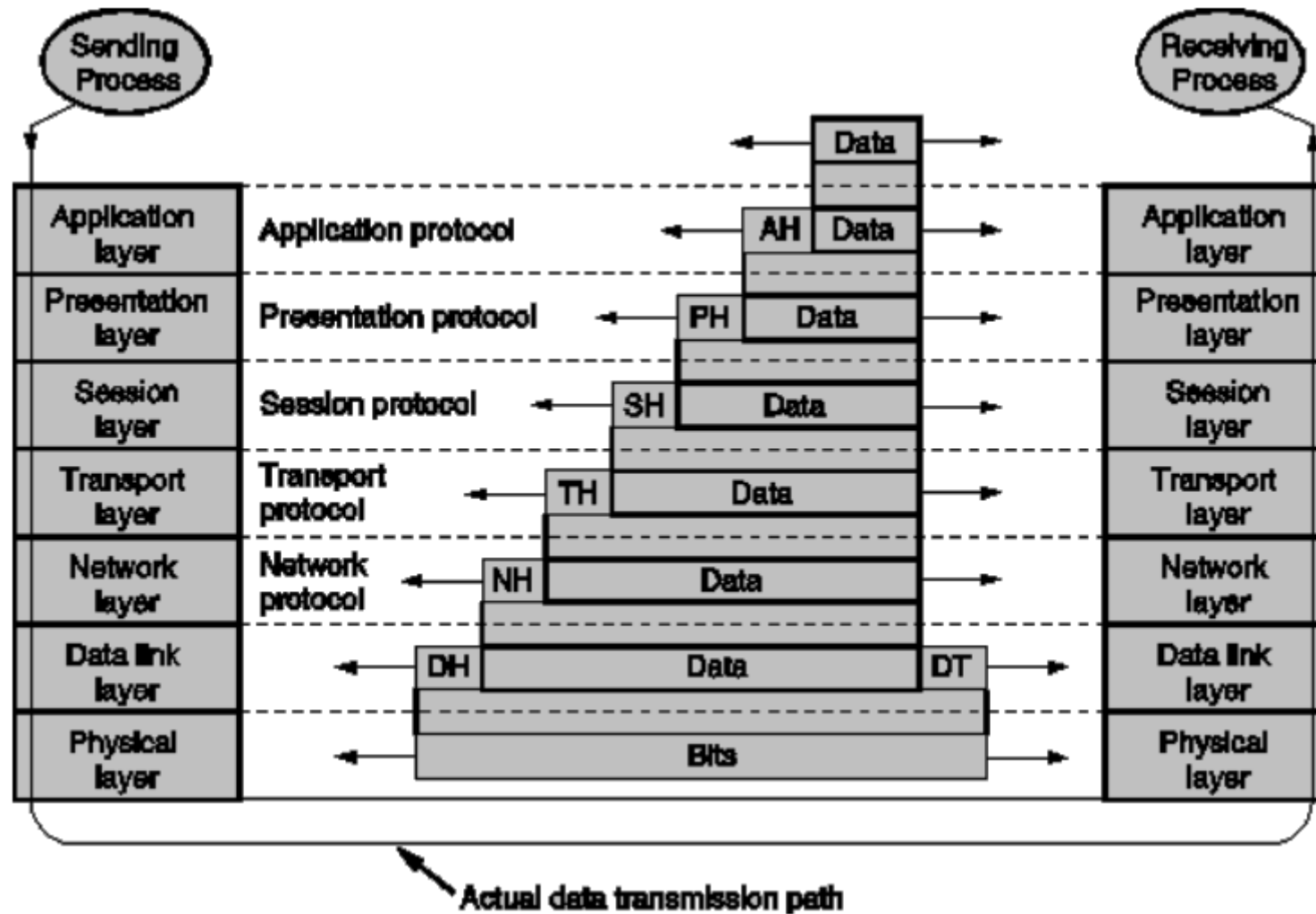


Data Link layer –
Bridge, Layer-2-Switch, Access Point



Physical layer –
network cable, Repeater, Hub

Data Transmission



Recap

- Peer-Peer
- Hub
- Switch
- Router
- ISP
- IAD

Useful links to recap at home:

https://www.youtube.com/watch?v=1z0ULvg_pW8&t=23s (7'18'')

Recap

- MAC Address
- IP Address
- Network-ID / Host-ID
- Port

Useful links to recap at home:

<https://www.youtube.com/watch?v=ddM9AcreVqY> (14'48'')

<https://www.youtube.com/watch?v=XQ3T14SIIV4> (20'17'')

<https://www.youtube.com/watch?v=4xlzlgYGqW8&list=PLI9NdZbdtA0wK8OlgttkScKRxcMkvoev> &index=3 (10'58'')

Recap



Useful links to recap at home:

<https://www.youtube.com/watch?v=HAVcXPI7oUY> (47'02")